

Endogenous Trade Fragmentation

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Abstract

Firms that leave cross-bloc supply relationships reduce the trade surplus that deters disruption. Their exits can therefore raise the hazard faced by firms that remain. I embed this mechanism in a multisector sourcing model with route thinning and endogenous friendly capacity. The local equilibrium response is $(I - J)^{-1}\mathbf{b}$, where J is the Jacobian of sector best responses. A scalar fragmentation multiplier is exact only when sectors are decoupled. Nondegenerate folds can generate sectoral jumps and history dependence. In the reference scenario, the Jacobian has spectral radius 0.196 and the critical-sector response is amplified by 1.24. A temporary high-hazard episode selects the high-fragmentation branch only when the state crosses the unstable equilibrium. The corrective maintenance subsidy is positive when the government's value of independent sourcing lies below the deterrence externality. All quantitative fold distances and policy wedges are scenario results to be disciplined by public trade, input-output, and sanctions data.

JEL codes: F13, F51, F52, D85, C62.

Keywords: fragmentation, friend-shoring, supply chains, strategic complementarity, tipping points, deterrence.

1 Introduction

China supplied about 22 percent of United States goods imports in 2017 and about 14 percent in 2023. Total imports stayed high over those years. The gap moved to Vietnam, Mexico, and other countries positioned between the blocs (Alfaro and Chor, 2023; Freund et al., 2023). The movement was uneven: trade between geopolitical blocs has slowed relative to trade within them since Russia invaded Ukraine in February 2022, with the sharpest breaks in strategically sensitive products (Gopinath et al., 2025). The reallocation is a rewiring of the network along political lines, concentrated exactly where substitution is hardest.

The question of this paper is whether that rewiring is self-limiting or self-reinforcing: the smooth repricing of a one-time increase in geopolitical risk, or the early stage of an unraveling in which each firm's exit makes exit more attractive for the next. The distinction matters for more than forecasting. Governments are committing tens of billions of dollars to friend-shoring subsidies on the theory that private firms undervalue resilience, and the quantitative literature

prices fragmentation counterfactuals conditional on an imposed split, with costs ranging from 0.2 to 7 percent of global GDP by depth (Aiyar et al., 2023; Baqaee et al., 2024; Attinasi et al., 2023). Whether the split is something that happens *to* the trade network or something the network does to itself changes both the forecast and the sign of good policy. Existing work treats the risk as exogenous. The contribution here is to make it an equilibrium object.

The model is deliberately spare. In each sector, firms choose between a cheap cross-bloc supplier exposed to disruption and a politically aligned supplier charging an idiosyncratic premium. Escalation destroys surplus on the trade that remains. A government weighs that loss against the political benefit of disruption, so each private exit can weaken deterrence for other firms. Route thinning raises disruption losses as trade leaves. Expansion of aligned capacity lowers future switching premia. The model keeps these mechanisms separate because they imply different empirical moments and policy wedges.

Strategic complementarity is a vector object. At a locally stable equilibrium, the response to a hazard shifter is $(I - J)^{-1}\mathbf{b}$. Within-sector comparisons can recover direct threshold responses while leaving J unidentified. Cross-sector comparisons do not generally remove network effects because sectors have different premium distributions and aggregate loadings. The empirical design therefore requires state variation and cross-sector spillover moments.

Nonlinear amplification can produce a fold in a critical sector. The theorem requires tangency, transversality, and nonzero curvature. A maximum best-response slope above one is not sufficient. The reference composite map satisfies the fold conditions numerically and yields a band from 11.4 to 13.1 percent. This is a scenario property, not an estimate of the current state. A temporary hazard increase creates permanent fragmentation only if it moves the state across the unstable branch.

The policy wedge compares the hazard and route externalities imposed on firms that stay with the capacity benefit delivered to firms that switch. A government value v_m for independent sourcing subtracts from this wedge. The reference value of the maintenance wedge is 6.4 percent before v_m and instrument costs. The policy result is therefore a threshold $v_m < \tau_m^*$ rather than an unconditional recommendation against friend-shoring.

The resilience literature studies sourcing against exogenous disruption risk (Grossman et al., 2023). Network models generate discontinuous fragility through endogenous links (Elliott et al., 2022). Trade-and-conflict evidence establishes that commerce can deter escalation (Martin et al., 2008; Jackson and Nei, 2015). The contribution here is to place that deterrence relation inside firms' sourcing decisions and to characterize the resulting multisector equilibrium.

Recent measurement documents reallocation along political lines and its concentration in sensitive sectors (Gopinath et al., 2025; Fernández-Villaverde et al., 2024; Alfaro and Chor, 2023). Those facts motivate the model but do not identify its feedback parameters. Section 6 states the additional moments required to estimate them.

The quantitative exercise reports conditional phase diagrams. Welfare accounting includes switching premia and expected disruption losses. The independence value remains a policy input. Public-data moments are linked to every parameter that controls the Jacobian or the fold

conditions.

2 The Model

The analysis separates general sign results from affine-hazard calculations. Strategic complementarity follows from the escalation-benefit density and the slope of destroyed trade surplus. The quantitative section uses a uniform benefit distribution to obtain an affine hazard. Fold existence and global branch structure are then verified numerically rather than inferred from a local slope alone.

2.1 Firms and the sourcing choice

There are two blocs and M sectors. In sector m , a unit mass of Home firms currently source a key intermediate input from a cross-bloc supplier, and sector m carries a share w_m of cross-bloc intermediate trade. Time is discrete and annual. Each firm chooses a sourcing mode: *cross-bloc*, keeping the incumbent supplier, or *friend-shored*, replacing it with an in-bloc supplier of the same input.

Friend-shoring costs firm i a flow premium g_i , expressed ad valorem on the input: the in-bloc supplier is dearer or must first be built. The premium is idiosyncratic, $g_i \sim H_m$, lognormal with median $\bar{g}_m$ and log dispersion s . Heterogeneity stands in for relationship capital and product-specific reshoring costs. A fraction f_m^0 of firms switched before geopolitics entered the calculus, for wages or diversification, and I take that margin as given.

Cross-bloc sourcing is cheap but exposed. With hazard π_m per year the link is severed by policy: an export control or an embargo. A severed firm loses λ_m , the disruption loss expressed as a flow-equivalent ad valorem cost, and then friend-shores perforce. The loss has a standard CES anatomy. If the firm's input bundle aggregates the cross-bloc variety and substitutes with elasticity σ_m , and the variety carries cost share s_m , losing it raises the input cost index by the factor $(1 - s_m)^{1/(1-\sigma_m)}$, so a disruption lasting D_m years costs

$$\lambda_m^0 = \left[(1 - s_m)^{\frac{1}{1-\sigma_m}} - 1 \right] D_m, \quad (1)$$

large precisely where substitution is hard (σ_m low), exposure deep (s_m high), and recovery slow (D_m long). Equation (1) is what makes “criticality” a parameter rather than an adjective, and it is measurable sector by sector from substitution elasticities and input-output tables (Broda and Weinstein, 2006; OECD, 2026). The disruption studies of Barrot and Sauvagnat (2016), Boehm et al. (2019), and Carvalho et al. (2021) document both the losses and recovery times on the order of a year for specific inputs.

The sourcing choice reduces exactly to a threshold rule.

Lemma 1 (The flow comparison is exact). *Suppose a disrupted firm switches perforce and firms discount at any $\beta_f \in (0, 1)$. In a stationary environment, staying is optimal if and only if the*

flow comparison holds:

$$\pi_m \lambda_m \leq g_i^{\text{eff}},$$

where g_i^{eff} is the effective premium defined below. Forward-looking and myopic thresholds coincide.

The logic is that switching is an absorbing state whose cost is the premium forever, while staying pays the expected disruption loss and, upon disruption, inherits the same absorbing state, so the continuation values cancel and the flow comparison survives intact (Appendix A). Along transition paths a small option-value correction appears. Appendix C bounds it and connects the selection issue to Krugman (1991) and Matsuyama (1991).

2.2 The feedback, derived

Let $F_m \in [0, 1]$ denote the fragmented share of sector m 's cross-bloc trade, and let $F = \sum_m w_m F_m$.

The hazard. Disputes arrive with probability δ_m . Conditional on a dispute, escalation yields benefit b with distribution G_m and density g_m^b . It destroys surplus

$$\mathcal{C}_m(F_m, F) = \vartheta_m \bar{S}_m(1 - F_m) + \vartheta_a \bar{S}(1 - F). \quad (2)$$

The government escalates when $b \geq \mathcal{C}_m$. An additional systemic component $\eta_a F$ captures disruptions originating outside the sector-specific dispute. The hazard is

$$\pi_m(F_m, F) = \delta_m [1 - G_m(\mathcal{C}_m(F_m, F))] + \eta_a F. \quad (3)$$

Under a uniform benefit distribution and an interior escalation probability, the quantitative specification sets $\vartheta_a = 0$ in $\mathcal{C}_m$ and retains cross-sector exposure through $\eta_a F$. Normalization of the intercept gives

$$\pi_m(F_m, F) = \delta_m [1 - \theta(1 - F_m)] + \eta_a F, \quad (4)$$

where θ is the sector deterrence share. The affine equation is used in the quantitative exercise. The sign results use (3). In particular,

$$\frac{\partial \pi_m}{\partial F_n} = -\delta_m g_m^b(\mathcal{C}_m) \frac{\partial \mathcal{C}_m}{\partial F_n} + \eta_a w_n > 0 \quad (5)$$

whenever remaining trade raises the cost of escalation. Uniformity is required for linearity, not for strategic complementarity.

The loss: routes thin as trade leaves. Shared infrastructure spreads disruption: alternative routings, insurance pools, spot markets, and buffer inventories are thick when many firms use the corridor and thin when few do. I let the loss rise linearly with the sector's fragmentation,

$$\lambda_m(F_m) = \lambda_m^0 (1 + \xi F_m), \quad (6)$$

with $\xi \geq 0$ the thinning elasticity.

The premium: friendly capacity scales. The in-bloc supply base gets cheaper as it gets bigger: entry, learning, and fixed-cost spreading in the destination countries compress the premium. The effective premium is

$$g_i^{\text{eff}} = g_i (1 - \chi F_m), \quad (7)$$

with $\chi \geq 0$ the capacity-scale elasticity. This is the channel the post-2018 reallocation to Vietnam and Mexico makes visible (Alfaro and Chor, 2023): each cohort of switchers builds the capacity that lowers the next cohort's cost.

Combining (4)–(7) with Lemma 1, firm i friend-shores when g_i falls below the threshold

$$\hat{g}_m(F_m, F) = \frac{\pi_m(F_m, F) \lambda_m^0 (1 + \xi F_m)}{1 - \chi F_m}, \quad (8)$$

and the sector's fragmented share consistent with individual choices is

$$F_m = T_m(F_m, F) \equiv f_m^0 + (1 - f_m^0) H_m(\hat{g}_m(F_m, F)). \quad (9)$$

Proposition 1 (Strategic complementarity). *Suppose $g_m^b(C_m) > 0$, $h_m(\hat{g}_m) > 0$, $\partial C_m / \partial F_n < 0$, and $1 - \chi F_m > 0$. Then T_m is increasing in every F_n . Under the affine hazard, the own-sector threshold derivative is*

$$\frac{\partial \hat{g}_m}{\partial F_m} = \underbrace{\frac{\delta_m \theta \lambda_m^0 (1 + \xi F_m)}{1 - \chi F_m}}_{\text{deterrence erosion}} + \underbrace{\frac{\pi_m \lambda_m^0 \xi}{1 - \chi F_m}}_{\text{route thinning}} + \underbrace{\frac{\pi_m \lambda_m^0 (1 + \xi F_m) \chi}{(1 - \chi F_m)^2}}_{\text{capacity scale}} > 0,$$

The cross-sector derivative is

$$\frac{\partial \hat{g}_m}{\partial F_n} = \frac{\eta_a w_n \lambda_m^0 (1 + \xi F_m)}{1 - \chi F_m} > 0$$

when $\eta_a > 0$. If $\theta = \xi = \chi = \eta_a = 0$, the threshold is independent of fragmentation and the equilibrium is unique.

Deterrence erosion is a security externality. Route thinning and capacity scale are market externalities of the kind studied by Cooper and John (1988). Their relative size determines whether a fold exists and which policy margin is corrective.

The static objects are now in place: a threshold economy in each sector, coupled through the hazard and the aggregate. What remains is to characterize the fixed points of (9), and that is where the multiplier and the tipping structure live.

3 Equilibrium: Multipliers, Tipping, Hysteresis

3.1 Structure and the fragmentation multiplier

An equilibrium is a vector $\mathbf{F} = (F_1, \dots, F_M)$ satisfying $\mathbf{F} = \mathbf{T}(\mathbf{F}, z)$. Let $J = D_{\mathbf{F}}\mathbf{T}$ denote the Jacobian, with element $J_{mn} = \partial T_m / \partial F_n$.

Proposition 2 (Multisector equilibrium response). *The continuous map $\mathbf{T}$ has an equilibrium on $\prod_m [f_m^0, 1]$. At an equilibrium with spectral radius $\varrho(J) < 1$, the local response to a scalar primitive z is*

$$\frac{d\mathbf{F}^*}{dz} = (I - J)^{-1} \mathbf{b}_z, \quad \mathbf{b}_z \equiv D_z \mathbf{T}. \quad (10)$$

All elements of $(I - J)^{-1}$ are nonnegative. If the system is sectorally decoupled, (10) reduces to

$$\frac{dF_m^*}{dz} = \frac{b_{mz}}{1 - J_{mm}}.$$

The scalar formula is exact only in that special case.

Corollary 1 (Direct effects and network effects). *Within-sector comparisons that hold $\mathbf{F}$ fixed identify components of $\mathbf{b}_z$. They do not identify J . Cross-sector comparisons generally retain network terms because sectors differ in H_m , λ_m^0 , and their loadings on aggregate fragmentation. Identification of $(I - J)^{-1}$ requires variation that moves sector states together with moments on cross-sector spillovers.*

At the reference scenario, $\varrho(J) = 0.196$. The critical-sector direct derivative with respect to δ is 0.885 and its total derivative is 1.101. The corresponding middle-sector derivatives are both approximately 1.2×10^{-7} . The critical-sector response is amplified by 1.24 and the trade-weighted aggregate response by 1.24. Holding the systemic term fixed gives an own-sector amplification of 1.24. Cross-sector spillovers account for 1.9 percent of the critical-sector amplification. The systemic channel is positive but quantitatively small in this calibration.

3.2 Folds and the sectoral cascade

Proposition 3 (Nondegenerate folds). *Let the remaining sectors lie on locally unique branches and write the resulting scalar residual as $R_m(F_m, \delta) = \tilde{T}_m(F_m, \delta) - F_m$. A nondegenerate fold at (F_m^*, δ^*) satisfies*

$$R_m = 0, \quad R_{mF} = 0, \quad R_{mFF} \neq 0, \quad R_{m\delta} \neq 0. \quad (11)$$

Near this point, two equilibria exist on one side of δ^* and none exist on the other. A hysteresis band requires two such folds together with the boundary signs that generate a low and a high branch. If $\eta_a = 0$ and sectors differ only in λ_m^0 , fold hazards satisfy $\delta_m^* \lambda_m^0 = K$ for a common constant K . An isolated sector jump changes aggregate fragmentation by $w_m \Delta F_m$.

The fold theorem is local. The reported global band is a numerical property of the calibrated composite map. The solver verifies two changes in the number of transversal roots and checks tangency at their limiting locations. Premium dispersion determines whether the best response becomes steep enough to meet the nondegeneracy conditions. At log dispersion 0.5, the critical-sector band is [11.4, 13.1] percent. Raising dispersion to 0.7 removes the band in the reported parameter range.

A sector carrying fifteen percent of cross-bloc trade can jump while the aggregate moves by only its trade weight times the sectoral change. Distance to instability is measured by $\varrho(J)$ in the full system. A scalar sector slope is adequate only when off-diagonal entries are negligible.

At a high-fragmentation equilibrium the hazard is higher by $\delta_m \theta \Delta F_m + \eta_a \Delta F$. Firms respond to disruption risk, and their response raises that risk when remaining trade deters escalation.

3.3 Dynamics and hysteresis

Relationships are sticky: a fraction ρ of firms reconsider their sourcing each year, so the sector state evolves as

$$F_{m,t+1} = (1 - \rho) F_{m,t} + \rho T_m(F_{m,t}, F_t, \delta_t), \quad (12)$$

a monotone map in both the state and the hazard. Nothing locks a firm in: a reconsidering firm returns to cross-bloc sourcing whenever the threshold condition says to.

Proposition 4 (Temporary shocks, permanent networks). *Let the hazard follow a spike path: $\delta_t = \delta^{sp} > \delta_{hi}^*$ for T^{sp} periods, then $\delta_t = \delta^{term} \in (\delta_{lo}^*, \delta_{hi}^*)$ forever. There is a finite duration $\bar{T}(\delta^{sp}, \rho)$, decreasing in the overshoot and in the reconsideration rate, such that the economy converges to the high-fragmentation equilibrium of δ^{term} if $T^{sp} \geq \bar{T}$, and returns to the low one if $T^{sp} < \bar{T}$. Reversal from the high branch requires the hazard to fall below δ_{lo}^* , while the band $[\delta_{lo}^*, \delta_{hi}^*]$ is a region of history dependence.*

The proposition is a result about the partial-adjustment rule in (12). Static multiplicity means that identical fundamentals admit several sourcing equilibria. Dynamic hysteresis here means that the predetermined state crosses the unstable fixed point. Expectational selection allows forward-looking firms to coordinate on a branch without that crossing. Physical irreversibility would arise from sunk investment or relationship capital and is absent from the baseline. Appendix C explains why these mechanisms need not select the same path.

The model is now complete as positive economics. Its normative content, who should pay whom at the margin, turns out to depend on where the sector sits relative to the fold, and that is the next section.

4 Welfare and Policy

4.1 The planner's margin and the sign of the wedge

Take a planner who can choose the sector's fragmented share, implementing it with threshold allocations, and who counts resource costs: premia actually paid, and disruption losses actually

expected. Per unit of sector trade value,

$$B_m(F) = (1 - f_m^0)(1 - \chi F) \int_0^{q(F)} g dH_m + (1 - F) \pi_m(F) \lambda_m(F), \quad (13)$$

where $q(F)$ is the premium of the marginal switcher. Differentiating and evaluating at the decentralized equilibrium, where the marginal switcher is privately indifferent, the private terms cancel and what remains is the net external effect of one more exit:

$$\tau_m^*(F) = \underbrace{(1 - F)[\lambda_m(F) \delta_m \theta + \pi_m \lambda_m^0 \xi]}_{\text{hazard and loss imposed on stayers}} - \underbrace{\chi (1 - f_m^0) \int_0^{q(F)} g dH_m}_{\text{premium relief delivered to switchers}}. \quad (14)$$

Proposition 5 (The wedge and the independence value). *$\tau_m^*(F) > 0$ at $F = f_m^0$ whenever $\theta > 0$. If $\chi > 0$, then $\tau_m^*(F) < 0$ as $F \rightarrow 1$, so a sign change exists. Let v_m be the government's flow value of one more unit of independent sourcing. The net wedge on exit is $\tau_m^*(F) - v_m$. A maintenance subsidy is corrective if and only if $v_m < \tau_m^*(F)$. Under the single-crossing condition verified for the reference scenario, the zero of τ_m^* is unique.*

An exit raises the disruption hazard on remaining links and thins the routes available to firms that stay. It also expands friendly capacity. The former forces make decentralized exit excessive. The capacity effect works in the opposite direction. In the reference scenario, τ_m^* is +6.4 percent at $\delta = 0.10$ and -4.9 percent at $\delta = 0.24$. These are conditional policy wedges before subtracting v_m .

4.2 Instruments

Policy can act on the switching premium, the disruption loss, or the return to maintaining trade.

A *friend-shoring subsidy* pays share ς of the switching premium. It shifts the threshold to $\hat{g}_m/(1 - \varsigma)$: mechanically it raises F_m , erodes the hostage, raises π_m , and drags both fold points toward the economy. It is the corrective instrument only where $\tau^* < 0$, past the flip. Deployed below the fold it has the wrong sign, and deployed near the fold it can tip the sector outright, a possibility Section 5 prices at the calibration.

A *stockpile or buffer* cuts the loss a disruption inflicts: λ_m^0 falls by the covered fraction. It lowers thresholds, lowers F_m , preserves deterrence, and pushes the folds away. In the model's accounting it is the main resilience instrument because it purchases exactly the object firms fear, disruption cost, while preserving the object society needs, the hostage. Its resource cost prices the comparison in practice.

A *maintenance subsidy* of τ^* on cross-bloc trade in critical inputs implements the planner's optimum directly. Its political economy is hard, subsidizing trade with the adversary in the name of security, and the model supplies the logic: the subsidy buys deterrence, which is a public good priced at zero by every private contract.

The independence value v_m is not calibrated from observed subsidies. Proposition 5 instead reports the threshold. At the reference state, a zero maintenance subsidy requires $v_m \geq 6.4$

Table 1: Calibration (annual).

Parameter		Value	Category	Basis
Trade weights	w_m	0.15 / 0.35 / 0.50	scenario types	critical / middle / flexible
Substitution elasticities	σ_m	2 / 4 / 8	external range	HS-level estimates (BW)
Cross-bloc input shares	s_m	0.50 / 0.35 / 0.20	scenario types	OECD ICIO moments
Disruption durations	D_m	1.5 / 1.0 / 0.5 yr	external range	published case studies
Implied losses	λ_m^0	1.50 / 15% / 1.6%	model implied	equation (1)
Median premia	$\bar{g}_m$	0.20 / 0.15 / 0.10	scenario	premium-design target
Premium dispersion	s	0.5	scenario	0.3 to 1.1 sensitivity
Pre-existing diversification	f_m^0	0.05 / 0.08 / 0.10	scenario	pre-period trade shares
Deterred share	θ	0.6 {0.4–0.7}	scenario range	sign from MMT
Thinning, capacity	ξ, χ	0.4, 0.4 {0–0.6}	scenario range	empirical targets
Systemic feedback	η_a	0.01	scenario	reported decomposition
Reconsideration rate	ρ	0.25	external proxy	four-year links (BS)
Low dispute hazard	δ^{pre}	4%	scenario	full axis reported
High dispute hazard	δ^{post}	10%	scenario	full axis reported
Cross-bloc intermediates / GDP	α	4%	public aggregate	trade and IO data

Notes: BW is Broda and Weinstein (2006). MMT is Martin et al. (2008). BS is Barrot and Sauvagnat (2016). Braces give scenario ranges. Values labeled scenario are not estimates.

percent of sector trade value per year. This inequality is the policy result. The model does not assign a welfare value to independence.

5 Quantitative Analysis

5.1 Calibration

Table 1 separates external evidence from model-implied objects and scenario inputs. The calculation is a reference scenario rather than an estimate of the world economy’s distance to a fold.

Equation (1) maps the external elasticity and duration ranges into disruption losses. The premium distribution and feedback elasticities determine whether the composite best response develops a fold. Their baseline values are scenarios to be replaced by the moments in Section 6. The change from $\delta = 4$ percent to $\delta = 10$ percent is a reference reassessment. It is not an estimate of beliefs before and after 2022.

5.2 Reference multipliers and fold distance

The high-hazard reference equilibrium puts critical-sector fragmentation at 6.9 percent, against 5.0 percent in the low-hazard scenario. The aggregate moves from 8.6 to 8.8 percent. The exercise illustrates how heterogeneous criticality concentrates adjustment. It does not attribute observed trade reallocation to the 2022 shock.

The calibrated composite map has a critical-sector band from 11.4 to 13.1 percent. Moving between the two reference hazards covers 66 percent of the distance from the low scenario to the upper fold. The sensitivity grid produces distances from 47.0 to 100.0 percent. Some cells

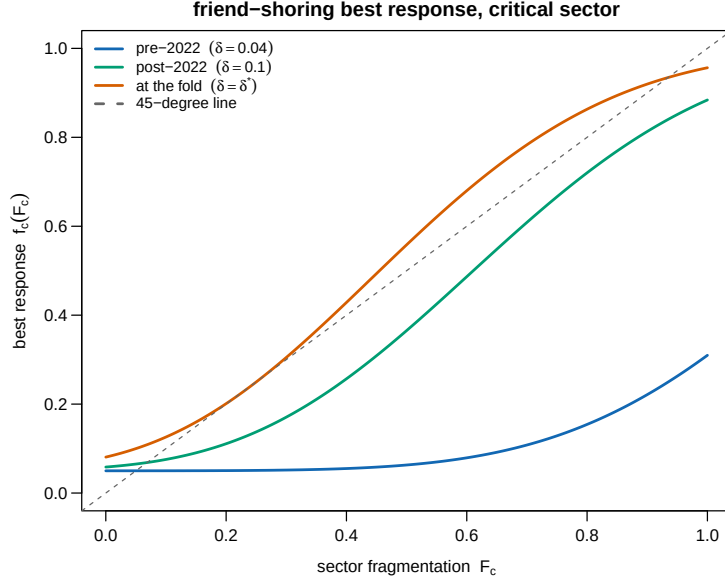


Figure 1: The critical sector’s best response at selected dispute hazards. At the fold, the curve is tangent to the diagonal and the low equilibrium disappears.

have no fold and one cell has already crossed it at the high reference hazard. These outcomes show that fold distance is identified only after the deterrence, capacity, and premium-distribution moments have been estimated.

5.3 Hysteresis: a temporary confrontation, a permanent network

Figure 3 is a conditional phase experiment. Both economies end at $\delta = 12.2$ percent. One reaches that value directly and converges to 12 percent fragmentation. The other spends twelve years at $\delta = 20.1$ percent and converges to 90 percent after the hazard returns. The difference follows from crossing the unstable equilibrium. It is not a forecast for a twelve-year geopolitical episode. The associated resource-cost difference is 4.4 percent of sector trade value under the model’s restricted accounting.

5.4 Policy at the calibration

In the reference scenario, a subsidy covering 25 percent of the switching premium moves the upper fold to 9.7 percent and selects a high-fragmentation equilibrium at the high hazard. A stockpile that reduces disruption loss by one third moves the upper fold to 19.7 percent. These rankings are conditional on v_m , the cost of the instruments, and the estimated deterrence relation. Figure 5 reports the model wedge before those additional terms are included.

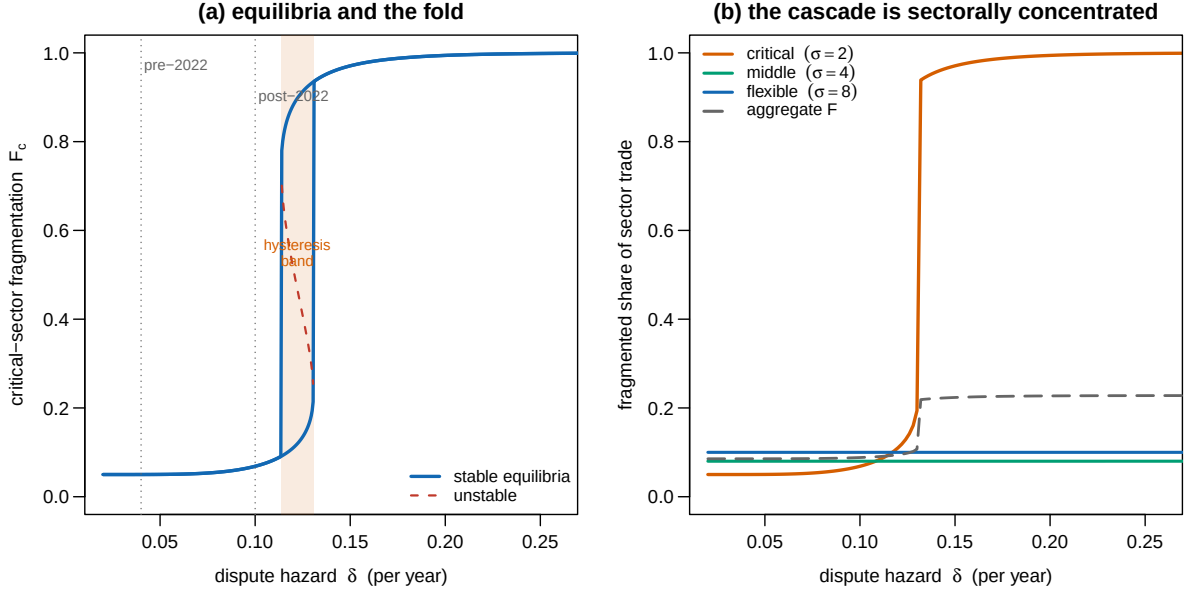


Figure 2: Panel (a): equilibria of the critical sector by dispute hazard. Solid branches are stable, dashed unstable, and the shaded region is the hysteresis band $[\delta_{lo}^*, \delta_{hi}^*] = [11.4\%, 13.1\%]$. Panel (b): fragmented shares by sector along the same axis, continuing from low initial conditions: the cascade is confined to the critical sector.

6 Empirical Designs

The empirical unit is an importer-exporter-HS6 relationship, not an unobserved firm-supplier link. Public data discipline the direct reallocation response and each component of the Jacobian separately. Distance to a fold is computed only after those moments are jointly estimated.

Direct reallocation. Product-level bilateral trade from BACI (CEPII, 2026a) at HS6 enters the specification

$$\text{TradeShare}_{ijpt} = \alpha_{ijp} + \alpha_{ipt} + \alpha_{jpt} + \beta \text{Post}_t \text{GeoDist}_{ij} \text{Criticality}_{ip} + \varepsilon_{ijpt}, \quad (15)$$

where all exposure variables are constructed before 2022. Criticality combines short-run substitution evidence, OECD ICIO input shares, and public import-dependence measures. The coefficient is descriptive because the 2022 period combines war, sanctions, commodity shocks, and industrial policy. The 2018 tariff episode is a comparison period rather than a placebo because it also induced persistent supply investment.

Relationship exit and re-entry. Define $\text{Link}_{ijpt} = \mathbf{1}\{\text{Trade}_{ijpt} > \underline{v}\}$ using a threshold stated before estimation. Exit and re-entry hazards are estimated at the importer-exporter-product level. Slow re-entry is consistent with partial-adjustment hysteresis, sunk relationship capital, or persistent expectations. It does not identify the model's basin mechanism by itself. The interaction with independently estimated fold proximity supplies the discriminating restriction.

Deterrence. GSDB alone does not record the full sequence of geopolitical disputes and

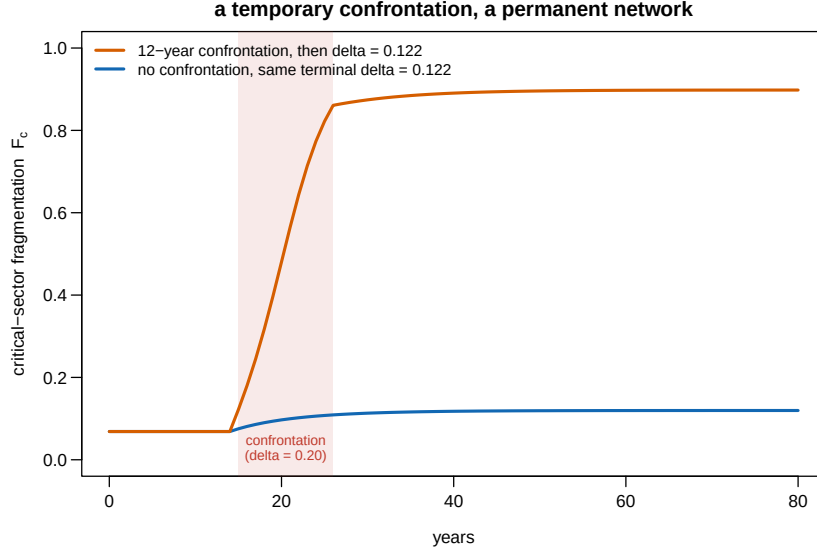


Figure 3: Two economies facing the identical terminal dispute hazard $\delta = 12.2\%$, inside the band. One lived through a twelve-year confrontation at $\delta = 20.1\%$. The other entered directly at the terminal hazard.

non-escalations. The deterrence equation is therefore estimated at the dyad-year level after conditioning on a public dispute chronology:

$$\Pr(\text{Escalation}_{ij,t} = 1 \mid \text{Dispute}_{ij,t} = 1) = \Lambda(\alpha_{ij} + \tau_t + \beta \widehat{\text{TradeExposure}}_{ij,t-1} + \Gamma' X_{ij,t-1}), \quad (16)$$

where trade exposure is predicted from gravity variables or historical determinants. Realized trade is endogenous to political relations. Conditioning on observed disputes aligns the regression with the model's separation between dispute arrival and escalation. Sector-specific deterrence is not claimed without a product-level export-control chronology.

Premiums and capacity. The premium g_{ijpt} is the within-importer-product difference between aligned-supplier and cross-bloc unit values. Importer-product-year effects absorb common demand. Exporter-product effects absorb persistent quality. Small quantities are trimmed and CEPII trade-price indices provide a robustness measure. The capacity elasticity comes from the response of aligned-supplier prices to cumulative reallocation. A shift-share instrument based on third-country demand is feasible, but its exclusion restriction fails if global product shocks move prices directly. Results are therefore reported with and without product-specific global controls.

Route thinning. IMF PortWatch disruptions link recovery time to pre-event corridor thickness. Thickness is measured by active exporter counts, shipment value, and alternative-route capacity. This design maps directly to ξ and does not infer recovery duration from unit values.

The estimated parameter vector is propagated jointly through the fixed-point solver. A bootstrap over the empirical moments reports the distribution of $\varrho(J)$ and the fold location. One-at-a-time sensitivity remains a diagnostic and is not the uncertainty statement.

The deterrence regression disciplines the sign and scale of the escalation externality. Relation-

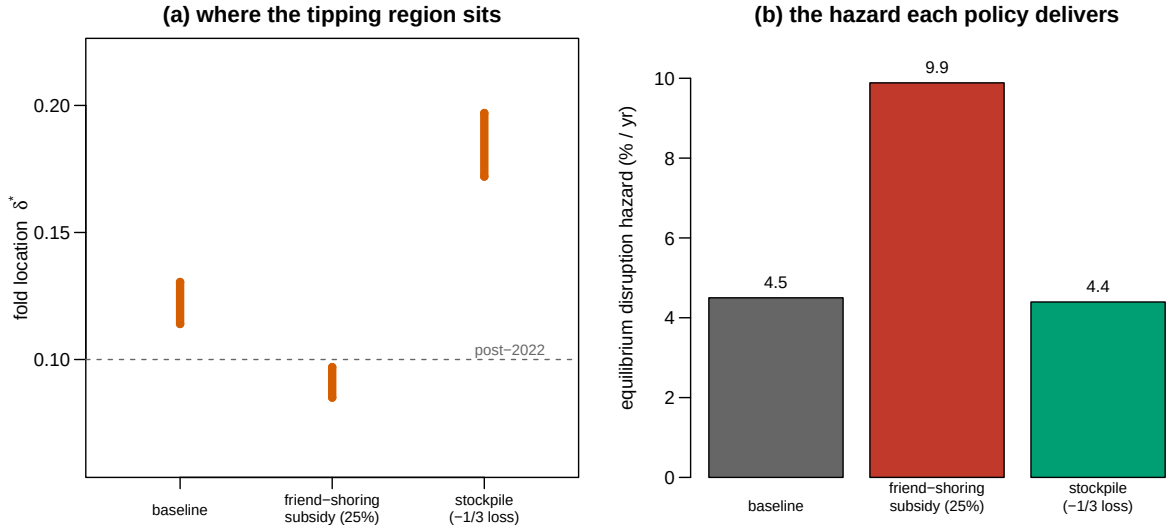


Figure 4: Policy scenarios. The dashed line is the high reference hazard. Panel (b) reports the equilibrium disruption hazard at $\delta = 10\%$.

ship persistence is informative only after sunk costs and expected future policy are controlled. The structural claim rests on their joint fit with the capacity and route-thinning moments.

Every dataset in the empirical program is public. Table 3 records the implementation.

7 Conclusion

Decentralized sourcing places the trade-conflict relation inside firms' private decisions. An exit reduces the surplus that escalation would destroy and can raise the disruption hazard on remaining links. In a multisector economy, the equilibrium amplification is governed by the Jacobian of best responses. Sectoral criticality alone does not determine the multiplier.

The fold result requires nondegenerate tangency and global crossing conditions. The reference scenario satisfies these conditions and displays history dependence under partial adjustment. It does not establish that the current economy is near a fold or that forward-looking firms select the same path. Those claims require the empirical moments specified in Section 6.

Policy depends on the government's value of independent sourcing. A maintenance subsidy is corrective when $v_m < \tau_m^*$. The deterrence and route externalities raise τ_m^* . Friendly capacity lowers it. This threshold states the tradeoff without treating an illustrative scenario as a universal prescription.

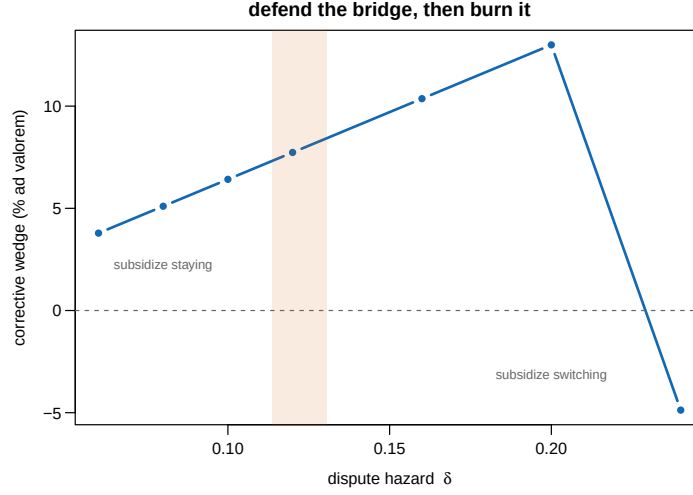


Figure 5: The corrective wedge τ^* by dispute hazard: positive (subsidize maintained cross-bloc trade) below and inside the band, negative (subsidize switching) beyond it.

Table 2: Predictions and implementation.

Prediction	Model magnitude	Null	Estimation
P1: cross-bloc shares fall with exposure	conditional sign	0	BACI triple difference
P2: response dose-monotone in constructed criticality λ_m^0	ordering of (1)	flat	criticality interactions
P3: re-entry slows near independently measured criticality	interaction sign	no interaction	relationship hazards
P4: escalation hazard falls with predicted trade exposure	$\beta < 0$	$\beta = 0$	dyad-dispute panel
P5: aligned-supplier premia fall with cumulative demand	$\chi > 0$	$\chi = 0$	unit-value gaps
P6: local persistence rises with the leading eigenvalue	$\varrho(J) \uparrow$	constant	sector dynamics

Notes: the table reports sign and interaction restrictions. Scenario magnitudes are not used as empirical targets unless separately identified.

Table 3: Data and implementation.

Dataset	Coverage	Frequency	Access	Observed or proxied	Measure
CEPII BACI	HS6 bilateral trade	annual	public	observed	in hand
CEPII GeoDep	dependency and concentration	annual	public	observed	in hand
CEPII MACMap	applied tariffs at HS6	periodic	public	observed	in hand
OECD ICIO	cost shares and centrality	annual	public	observed	in hand
HS-level elasticities	Broda and Weinstein (2006)	fixed	published	estimated, period	pre- in hand
IMF PortWatch	maritime disruptions	event	public	observed	in hand
GSDB release 4	through 2023	episode	public request	observed	obtainable
MID or UCDP events	dyad disputes	annual	public	observed	in hand
UN ideal points	all members	annual	public	observed	in hand
CEPII TradePrices	product prices	annual	public	price proxy	in hand

Notes: BACI unit values remain noisy because quality and composition vary. TradePrices provides the principal robustness series. Product-level escalation requires a separate official-list chronology and is not assumed available.

A Proofs

A.1 Proof of Lemma 1

Let $V^{sw} = -g^{\text{eff}}/(1 - \beta_f)$ be the value of the absorbing friend-shored state and V^{st} the value of staying, which pays the expected loss and transits to the absorbing state upon disruption: $V^{st} = -\pi\lambda + \beta_f[(1 - \pi)V^{st} + \pi V^{sw}]$. Solving, $V^{st}(1 - \beta_f(1 - \pi)) = -\pi\lambda + \beta_f\pi V^{sw}$, and $V^{st} \geq V^{sw}$ iff $-\pi\lambda + \beta_f\pi V^{sw} \geq (1 - \beta_f(1 - \pi))V^{sw}$ iff $-\pi\lambda \geq (1 - \beta_f)V^{sw} = -g^{\text{eff}}$, which is the flow comparison. No discount-rate term survives. ■

A.2 Deterrence microfoundation

Lemma 2 (Linear hazard from escalation choice). *Suppose disputes arrive with probability δ_m and the adversary escalates when $b \geq \mathcal{C}_m$. Then the dispute component of the link hazard is (3) without $\eta_a F$. It is increasing in each fragmentation state whenever $g_m^b(\mathcal{C}_m) > 0$ and remaining trade raises $\mathcal{C}_m$. If b is uniform and the escalation probability is interior, the hazard is affine.*

The escalation probability is $1 - G_m(\mathcal{C}_m)$. Differentiation gives $-g_m^b(\mathcal{C}_m)\mathcal{C}_{mF_n} > 0$. With a uniform distribution on $[0, \bar{b}]$, the interior probability is $1 - \mathcal{C}_m/\bar{b}$. Multiplication by δ_m and normalization of its intercept give the dispute component of (4). The separate term $\eta_a F$ preserves monotonicity. ■

A.3 Proof of Proposition 1

Under the affine hazard, write

$$\hat{g}_m(F_m, F) = \lambda_m^0 \frac{\delta_m[(1 - \theta) + \theta F_m] + \eta_a F}{1 - \chi F_m} (1 + \xi F_m).$$

Holding F fixed, the product and quotient rules give

$$\frac{\partial \hat{g}_m}{\partial F_m} = \lambda_m^0 \left[\frac{\delta_m \theta (1 + \xi F_m)}{1 - \chi F_m} + \frac{\xi \pi_m}{1 - \chi F_m} + \frac{\chi \pi_m (1 + \xi F_m)}{(1 - \chi F_m)^2} \right].$$

This is the own-sector formula in the proposition. Holding F_m fixed gives

$$\frac{\partial \hat{g}_m}{\partial F} = \frac{\eta_a \lambda_m^0 (1 + \xi F_m)}{1 - \chi F_m} > 0$$

when $\eta_a > 0$. Under the general hazard, the same sign follows from (5) and the positive density of H_m . If all feedback parameters vanish, T_m is independent of $\mathbf{F}$ and the fixed point is unique. ■

A.4 Proof of Proposition 2

Continuity of $\mathbf{T}$ on the compact convex product set gives existence by Brouwer's theorem. Differentiating $\mathbf{F}^* = \mathbf{T}(\mathbf{F}^*, z)$ gives $(I - J)d\mathbf{F}^*/dz = \mathbf{b}_z$. If $\varrho(J) < 1$, the inverse exists and

equals $\sum_{k=0}^{\infty} J^k$. Proposition 1 makes J nonnegative, so every term of the inverse is nonnegative. Setting off-diagonal entries to zero gives the scalar special case. ■

A.5 Proof of Corollary 1

Within a sector and equilibrium, variation in an idiosyncratic shifter moves individual thresholds while holding $\mathbf{F}$ fixed. Such variation identifies elements of $\mathbf{b}_z$. Variation across sectors also changes H_m , λ_m^0 , and aggregate loadings, so the network component need not cancel. Recovering J requires state variation or structural restrictions on these loadings. ■

A.6 Proof of Proposition 3

At a point satisfying (11), the second-order expansion is

$$0 = R_{m\delta}(\delta - \delta^*) + \frac{1}{2}R_{mFF}(F_m - F_m^*)^2 + o(|\delta - \delta^*| + |F_m - F_m^*|^2).$$

The nonzero coefficients imply two nearby roots on one side and none on the other. A global band follows only when two folds are present and the residual has the stated boundary signs. With $\eta_a = 0$ and common non-loss primitives, the threshold depends on δ and λ_m^0 through their product. Hence $\delta_m^* \lambda_m^0$ is constant across sectors. The aggregate jump follows from $F = \sum_m w_m F_m$. ■

A.7 Proof of Proposition 4

The map $F \mapsto (1 - \rho)F + \rho T(F, \delta)$ is increasing in F and δ , so paths from ordered initial conditions stay ordered, and with constant δ the path is monotone toward the nearest fixed point in its direction of motion. During the spike, $\delta^{sp} > \delta_{hi}^*$ implies a unique fixed point $F^{hi}(\delta^{sp})$ exceeding the terminal unstable point $U(\delta^{term})$. The path rises toward it and crosses $U(\delta^{term})$ at some finite $\bar{T}$, decreasing in δ^{sp} (which raises the map) and in ρ (which speeds it). If reversion occurs after $\bar{T}$, the state exceeds $U(\delta^{term})$, where $T(F, \delta^{term}) > F$ holds on (U, F^{hi}) : the path rises monotonically to the high branch. If before, the state is below U , where $T < F$: it falls back. From the high branch, δ must fall below δ_{lo}^* , where the high fixed point is destroyed, before any monotone path can leave its basin. ■

A.8 Proof of Proposition 5

Differentiate (13) in F , using $(1 - f_m^0)h_m(q)q'(F) = 1$. The premium term gives

$$\frac{d}{dF} \left[(1 - f_m^0)(1 - \chi F) \int_0^{q(F)} g dH_m \right] = (1 - \chi F)q(F) - \chi(1 - f_m^0) \int_0^{q(F)} g dH_m.$$

The exposure term gives

$$\frac{d}{dF} [(1 - F)\pi_m(F)\lambda_m(F)] = -\pi\lambda + (1 - F) [\lambda \delta_m \theta + \pi \lambda_m^0 \xi].$$

At the decentralized equilibrium the marginal switcher is indifferent, $(1 - \chi F)q(F) = \pi\lambda$, so the private terms cancel. Therefore

$$\frac{dB_m}{dF} = (1 - F) \left[\lambda \delta_m \theta + \pi \lambda_m^0 \xi \right] - \chi(1 - f_m^0) \int_0^{q(F)} g dH_m = \tau_m^*(F).$$

A positive τ_m^* means one more exit raises social burden, so decentralized exit is excessive and the corrective instrument subsidizes maintained trade. At $F = f_m^0$ the integral vanishes and $\tau^* = (1 - f_m^0)[\lambda\delta\theta + \pi\lambda^0\xi] > 0$ when $\theta > 0$. As $F \rightarrow 1$, the positive bracket vanishes while the capacity term remains negative when $\chi > 0$. Continuity gives a crossing. A government value $v_m F$ subtracts v_m from the marginal burden, giving the net wedge $\tau_m^* - v_m$. A payment on maintained trade shifts the private comparison by the same amount and implements the threshold allocation. ■

B Microfoundations

The CES loss. Let the firm's input bundle be CES over the cross-bloc variety and a composite substitute, elasticity σ_m , with the variety's cost share s_m at initial prices. Removing the variety multiplies the bundle's price index by $(1 - s_m)^{1/(1-\sigma_m)}$: with $\sigma_m = 2$ and $s_m = 1/2$, a doubling, and with $\sigma_m = 8$ and $s_m = 0.2$, a three percent rise. A disruption lasting D_m years, before replacement contracts are in place, costs the flow equivalent (1). Downstream propagation through input-output rounds scales λ_m^0 by a centrality factor (Baqaei and Farhi, 2019; Carvalho et al., 2021). The baseline reports the direct sectoral loss, and the centrality extension prices the multiplier separately.

The escalation game. Lemma 2 treats one dispute. The same argument applies to repeated disputes so long as the escalation choice is static ex post, which it is here because portfolios are predetermined within the period. A forward-looking adversary, for whom today's escalation accelerates tomorrow's fragmentation and erodes future hostage-taking capacity, adds a second time-inconsistency problem on the government side. The present specification keeps the demand side clean and puts all strategic complementarity in firms' sourcing choices.

Capacity scale. The compression factor $(1 - \chi F_m)$ stands for entry and learning in the friendly supply base. With an in-bloc supply curve of elasticity $1/\chi$ facing cumulative demand proportional to F_m , the marginal premium falls linearly to first order. The empirical design therefore loads price compression on cumulative reallocation rather than contemporaneous volume.

C History versus Expectations

The dynamics (12) are adaptive: reconsidering firms take the current state as their forecast. With forward-looking firms, a reconsidering firm compares present values along an anticipated

equilibrium path, and inside the band $[\delta_{lo}^*, \delta_{hi}^*]$ there can exist self-fulfilling paths to either branch: the overlap of Krugman (1991) and Matsuyama (1991). The scope for expectations to override history is governed by the race between discounting and adjustment: when the reconsideration rate is slow relative to the discount rate, the future path matters little to the marginal chooser and history determines the outcome, and when firms are patient and adjustment fast, an anticipated migration can pay for itself. Lemma 1 sharpens the comparison here. Because continuation values cancel in the stationary comparison, the option term along a transition is proportional to the anticipated *change* in thresholds over the firm’s expected commitment spell, of length $1/\rho$, so the adaptive benchmark is exact in stationary environments and errs along transitions by that term. A full perfect-foresight treatment, including whether announcements alone can spring the trap of Proposition 4, is the natural next paper. The policy analysis of Section 4 is conservative in the meantime because coordination instruments only add to the planner’s kit.

D Parameter Discipline and Sensitivity

Table 4 records parameter status and scenario ranges. The exogenous-risk benchmark sets $\theta = \xi = \chi = \eta_a = 0$. Fold existence varies jointly with the feedback block and premium dispersion. The table is a scenario audit, not a confidence region.

Table 4: Parameter discipline and sensitivity.

Parameter	Enters at	Units	Source	Range examined	Effect of the range
σ_m, s_m, D_m	eq. (1)	elasticity, share, years	Broda and Weinstein (2006), ICIO, recovery studies	spanned across sectors	fix λ_m^0 , hence tipping order
λ_m^0	eq. (8)	ad val. flow	implied by eq. (1)	1.6% to 150%	fold position scales as $1/\lambda_m^0$
$\bar{g}_m, s$	eq. (9)	ad val. pts	scenario, premium log design	s 0.3 to 1.1	fold absent above $s \approx 0.7$
f_m^0	eq. (9)	share	scenario, pre-period target	floor check	level floor only
θ	eq. (4)	share deterred	scenario, deterrence design	0.4 to 0.7	fold location and Jacobian
ξ, χ	eqs. (6), (7)	elasticities	scenario, separate designs	0 to 0.6	fold existence and policy wedge
η_a	eq. (4)	per year	scenario	0 to 0.03	off-diagonal spillovers
ρ	eq. (12)	per year	Barrot and Sauvagnat (2016)	timing check	timing of $\bar{T}$ only
$\delta^{pre}, \delta^{post}$	eq. (4)	per year	scenario	full axis in figures	positions reference states
v_m	Prop. 5	ad valorem flow	policy input	threshold reported	maintenance-subsidy sign
w_m, α	aggregation	shares	trade aggregates	scale check	scale GDP numbers

Notes: the empirical implementation must use short-run substitution estimates. Long-run trade elasticities enter only sensitivity. Joint uncertainty is propagated after the component moments are estimated.

E Data Appendix

Trade. CEPII BACI, HS-6 bilateral flows and unit values, 1995–2024 (CEPII, 2026a), with extensive-margin relationship panels constructed at exporter-importer-product level. Bloc and alignment measures from UN ideal points (Bailey et al., 2017), fixed at pre-2022 values.

Criticality. The baseline empirical construction uses short-run substitution estimates, OECD ICIO cost shares, CEPII GeoDep concentration, and event-level recovery durations from Port-Watch (OECD, 2026; CEPII, 2024; International Monetary Fund, 2023). The long-run elasticities in Broda and Weinstein (2006) provide a sensitivity bound rather than the preferred short-run input. Every component is dated before the event used for validation.

Disputes and escalations. GSDB release 4 supplies sanctions episodes through 2023 (Global Sanctions Data Base, 2025). A public MID or UCDP-type chronology supplies the dispute risk set. Escalation is coded at the dyad-year level. Product-level deterrence is reported only if an official-list chronology can be assembled and validated.

Prices and tariffs. BACI unit values are compared with CEPII TradePrices. MAcMap-HS6 supplies bilateral applied tariffs at the product level (CEPII, 2026c; CEPII, 2026b).

Fragmentation. $F_{m,t}$ is measured as the share of sector m ’s pre-2022 cross-bloc trade value in sustained relationship exits, from the BACI extensive margin: the empirical counterpart of the model’s state variable.

F Extensions

Connector countries. Let measured friend-shoring reroute content through politically intermediate countries. Connectors lower the observed premium and raise fragmentation of direct links. If rerouted trade still creates surplus that escalation would destroy, measured bilateral fragmentation overstates deterrence erosion. Direct-trade and value-added exposure measures distinguish these cases (Gopinath et al., 2025; Freund et al., 2023).

Two-sided fragmentation. Making both blocs’ firms choose adds the adversary’s exits to the hostage erosion. With symmetric parameters the sector map’s slope doubles where deterrence is concerned, moving folds closer. The one-sided model is thus conservative about tipping.

Correlated disruption and percolation. Here a severed link harms only its firm. In production networks link shocks propagate (Elliott et al., 2022; Baqaee and Farhi, 2019), making λ depend on the pattern as well as the level of fragmentation. The interaction of strategic hazard with percolation fragility is unexplored and, I suspect, reinforcing.

The vulnerability motive. If the government values rewired trade at v per unit beyond firms' losses, the wedge becomes $\tau^* - v$ and the planner's problem otherwise stands. All sign results survive for $v < \tau^*$, and the policy discussion in Section 4 prices the threshold.

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Table 5: Sensitivity: the critical sector’s fold band $[\delta_{lo}^*, \delta_{hi}^*]$ by deterrence share θ and capacity elasticity χ , in dispute-hazard units. “Unique” means a unique equilibrium at all hazards.

	$\chi = 0.0$	$\chi = 0.2$	$\chi = 0.4$	$\chi = 0.6$
$\theta = 0.4$	none	none	0.107–0.107	0.086–0.099
$\theta = 0.5$	none	none	0.113–0.117	0.087–0.110
$\theta = 0.6$	none	0.138–0.141	0.116–0.133	0.088–0.127
$\theta = 0.7$	0.165–0.171	0.144–0.163	0.119–0.157	0.090–0.151

Figure 6 evaluates 165 scenario cells. The existence boundary slopes downward because deterrence and capacity feedback substitute in the best-response slope. At $\theta = 0.6$, a fold appears when $\chi \geq 0.15$. The grid contains 48 cells with a unique branch, 117 with a fold above the high reference hazard, and 16 with a fold below it. These counts summarize the scenario grid and carry no probability interpretation.

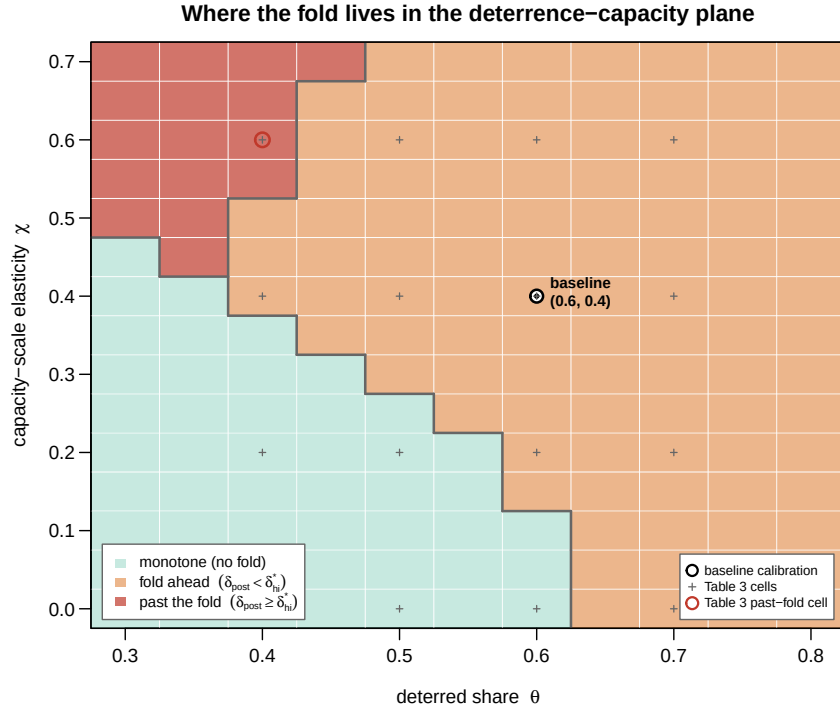


Figure 6: The fold in the (θ, χ) scenario plane. Shading classifies uniqueness, a fold above the high reference hazard, or a fold below it. Crosses mark the coarse-grid cells and the dot marks the reference scenario.